



EAST WEST UNIVERSITY

Course title: Object Oriented Programming

Course code: CSE110

Section: 08

Semester: Spring'26

Report Paper of Lab-01

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Covered Problems :

1. Ask user to enter marks and print the corresponding grade.
2. Write a Java program to check whether a triangle is Equilateral, Isosceles or Scalene.
3. Write a Java program to calculate the factorial of a given number.
4. Write a Java program to find a number is Armstrong or not.
5. Write a Java program to print the area and perimeter of a rectangle.
6. Write a Java program to swap two variables without using third variable.
7. Write a Java program to compute the sum of the first 100 prime numbers.

Instructor

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Question 1 : Ask user to enter marks and print the corresponding angle.

```
1  import java.util.Scanner;
2
3  public class lab1ex1 {
4      public static void main(String[] args) {
5
6          Scanner scanner = new Scanner(System.in);
7
8          System.out.println("Please enter your respective marks : ");
9          int marks = scanner.nextInt();
10         if (marks < 0) {
11             System.out.println("Invalid Marks");
12         } else if (0 <= marks && marks < 25) {
13             System.out.println("Your grade is : F");
14         } else if (marks >= 25 && marks < 45) {
15             System.out.println("Your grade is : E");
16         } else if (marks >= 45 && marks < 50) {
17             System.out.println("Your grade is : D");
18         } else if (marks >= 50 && marks < 60) {
19             System.out.println("Your grade is : C");
20         } else if (marks >= 60 && marks < 80) {
21             System.out.println("Your grade is : B");
22         } else {
23             System.out.println("Your grade is : A");
24         }
25     }
26 }
```

Output :

```
Please enter your respective marks :
45
Your grade is : D
```

Question 2 : Write a java program to check whether a triangle is Equilateral, Isosceles or Scalene.

```
1  import java.util.Scanner;
2
3  public class Lablex2 {
4      public static void main(String[] args) {
5
6          Scanner scanner = new Scanner(System.in);
7          System.out.println("Enter the value of side 1 : ");
8          int a = scanner.nextInt();
9          System.out.println("Enter the value of side 2 : ");
10         int b = scanner.nextInt();
11         System.out.println("Enter the value of side 3 : ");
12         int c = scanner.nextInt();
13
14         if (a == b && b == c && c == a) {
15             System.out.println("Your triangle is an Equilateral Triangle");
16         } else if (a == b || b == a || c == a) {
17             System.out.println("Your triangle is an Isosceles Triangle");
18         } else {
19             System.out.println("Your triangle is a Scalene Triangle");
20         }
21     }
22 }
23
24 }
```

Output :

```
Enter the value of side 1 :
60
Enter the value of side 2 :
60
Enter the value of side 3 :
50
Your triangle is an Isosceles Triangle
```

Question 3 : Write a Java program to calculate the factorial of a given number.

```
1  import java.util.Scanner;
2
3  public class Lab1ex3 {
4      public static void main(String[] args) {
5
6          Scanner scanner = new Scanner(System.in);
7          System.out.println("Enter the number to find out it's respective factorial : ");
8          int n= scanner.nextInt();
9          int factorial=1;
10         for (int i=1;i<=n;i++){
11             factorial = factorial*i;
12         }
13         System.out.println("The factorial of " + n + " will be : " + factorial);
14         scanner.close();
15
16     }
17 }
```

Output :

```
Enter the number to find out it's respective factorial :
5
The factorial of 5 will be : 120
```


Question 4 : Write a Java program to find a number is Armstrong or not.

```
1  import java.util.Scanner;
2  public class ArmstrongNumber {
3      static void main() {
4          Scanner scanner = new Scanner(System.in);
5          System.out.println("Enter the number : ");
6          int number = scanner.nextInt();
7          int originalNumber= number;
8          int digits=0;
9          int sum=0;
10         for (int temp = number; temp>0 ; temp= temp/10){
11             switch (digits++) {
12             }
13         }
14         for (int temp = number; temp>0 ; temp = temp/10){
15             int digit = temp % 10 ;
16             sum+=Math.pow(digit, digits);
17         }
18         if (sum==originalNumber){
19             System.out.println(originalNumber + " is an  Armstrong Number");
20         }
21         else{
22             System.out.println(originalNumber + " is not an Armstrong Number");
23         }
24         scanner.close();
25     }
26 }
```

Output :

```
Enter the number :
153
153 is an  Armstrong Number
```

Question 5 : Write a Java program to print the area and perimeter of a rectangle.

```
public class Lab1ex5 {  
    public static void main(String[] args) {  
        double width = 5.6;  
        double height = 8.5;  
  
        double area = width * height;  
        double perimeter = 2 * (width + height);  
  
        System.out.printf("Area is %.1f * %.1f = %.2f\n", width, height, area);  
        System.out.printf("Perimeter is 2 * (%.1f + %.1f) = %.2f\n", width, height, perimeter);  
    }  
}
```

Output :

Area is 5.6 * 8.5 = 47.60

Perimeter is 2 * (5.6 + 8.5) = 28.20

Question 6 : Write a Java program to swap two variables without using third variable.

```
1 ▶ public class Swap {  
2 ▶     public static void main(String[] args) {  
3         int a = 10;  
4         int b = 20;  
5  
6         a = a + b;  
7         b = a - b;  
8         a = a - b;  
9  
10        System.out.println("a = " + a);  
11        System.out.println("b = " + b);  
12    }  
13 }
```

Output :



```
Run [Swap] X  
C:\Program Files  
a = 20  
b = 10
```

Question 7 : Write a Java program to compute the sum of the first 100 prime numbers.

```
1 public class Lab1ex7 {  
2     public static void main(String[] args) {  
3         int sum = 0;  
4         int count = 0;  
5  
6         for (int number = 2; count < 100; number++) {  
7             boolean prime = true;  
8  
9             for (int i = 2; i <= number / 2; i++) {  
10                if (number % i == 0) {  
11                    prime = false;  
12                    break;  
13                }  
14            }  
15  
16            if (prime) {  
17                sum += number;  
18                count++;  
19            }  
20        }  
21  
22        System.out.println("Sum of the first 100 prime numbers: " + sum);  
23    }  
24 }
```

Output :

Sum of the first 100 prime numbers: 24133